Vectors I: Homework 1

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Introduction to Vectors in Three Dimensions. Original practice questions. Total: 50 marks. Give exact answers unless otherwise stated.

Questions

1

The vectors $\mathbf{a}$ and $\mathbf{b}$ are given by $\mathbf{a}=2\mathbf{i}-\mathbf{j}+3\mathbf{k}$ and $\mathbf{b}=-\mathbf{i}+4\mathbf{j}$.

(a) Write $\mathbf{a}$ and $\mathbf{b}$ as column vectors and find $\mathbf{a}+2\mathbf{b}$. **[2]**

(b) Find $\mathbf{c}$ such that $2\mathbf{a}-3\mathbf{b}+\mathbf{c}=\mathbf{0}$. **[2]**

(c) Points P and Q have position vectors $\mathbf{a}+2\mathbf{b}$ and $\mathbf{c}$ respectively. Find $\overrightarrow{PQ}$. **[1]**

2

The points A and B have coordinates $(1, -2, 2)$ and $(3, 1, -4)$ respectively.

(a) Find a unit vector that is parallel to AB. **[2]**

(b) Find a vector of length 21 parallel to AB. **[2]**

(c) A point P is such that $\overrightarrow{AP}$ has magnitude 14 and the same direction as $\overrightarrow{AB}$. Find the coordinates of P and state the order of the three points along their common straight path. **[2]**

3

The vectors $\mathbf{p}=\begin{pmatrix}6\\k\\-3\end{pmatrix}$ and $\mathbf{q}=\begin{pmatrix}2\\-4\\m\end{pmatrix}$ are parallel, where k and m are real constants.

(a) Find k and m. **[3]**

(b) State whether $\mathbf{p}$ and $\mathbf{q}$ have the same or opposite directions, and find $|\mathbf{p}|:|\mathbf{q}|$. **[1]**

(c) Find all vectors of magnitude $\sqrt{21}$ that are parallel to $\mathbf{p}$. **[2]**

4

The points A, B and C have coordinates $(1, 2, -1)$, $(3, 1, 2)$ and $(7, -1, 8)$ respectively.

(a) Show that A, B and C are collinear, and find $AB:BC$. **[3]**

(b) The point $R(u, -2, v)$ lies on the same straight path as A, B and C. Find u and v. **[3]**

5

The points A and B have coordinates $(2, -1, 3)$ and $(-4, 5, 0)$ respectively. A point M divides AB internally in the ratio $AM:MB=2:1$. A point N lies on AB produced beyond B and satisfies $AN:NB=5:2$.

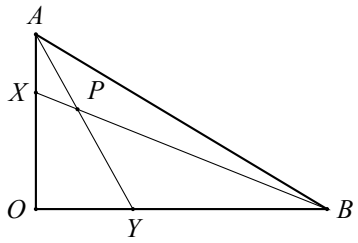
(a) Find the position vector of M. **[2]**

(b) Find the position vector of N. **[3]**

(c) State the order of the four points A, B, M and N, and find $MB:BN$. **[3]**

6

Relative to the origin O, the points A and B have non-zero, non-parallel position vectors **a** and **b** respectively. The point X lies on OA with $OX : XA = 2 : 1$, and the point Y lies on OB with $OY : YB = 1 : 2$. The line segments AY and BX intersect at P.



(a) By expressing $\overrightarrow{OP}$ in two ways using the ratio theorem, find $\overrightarrow{OP}$ in terms of **a** and **b**. **[6]**

(b) Find $AP : PY$ and $BP : PX$. **[2]**

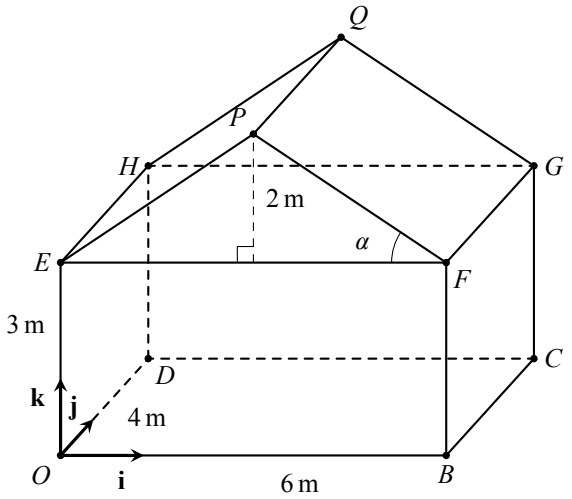
(c) The point Z divides AB internally with $AZ : ZB = 1 : 4$. Show that O, P, Z are collinear and find $OP : PZ$. **[2]**

7

A solid house consists of a cuboid $OBCDEFGH$ and a triangular prism roof $EFPHGQ$, as shown. The base $OBCD$ is horizontal. The vertices E, F, G and H are directly above O, B, C and D respectively.

The front-left corner O is the origin. The positive **i**, **j** and **k** directions are along OB, OD and OE respectively, and each unit represents 1 metre. It is given that $OB = 6$ m, $OD = 4$ m and $OE = 3$ m.

The ridge PQ is parallel to OD. The front gable EFP and back gable HGQ are congruent isosceles triangles, with P vertically above the midpoint of EF and Q vertically above the midpoint of HG. Each sloping roof face makes an angle α with the horizontal, where $\tan \alpha = \frac{2}{3}$; the vertical roof rise is 2 m.



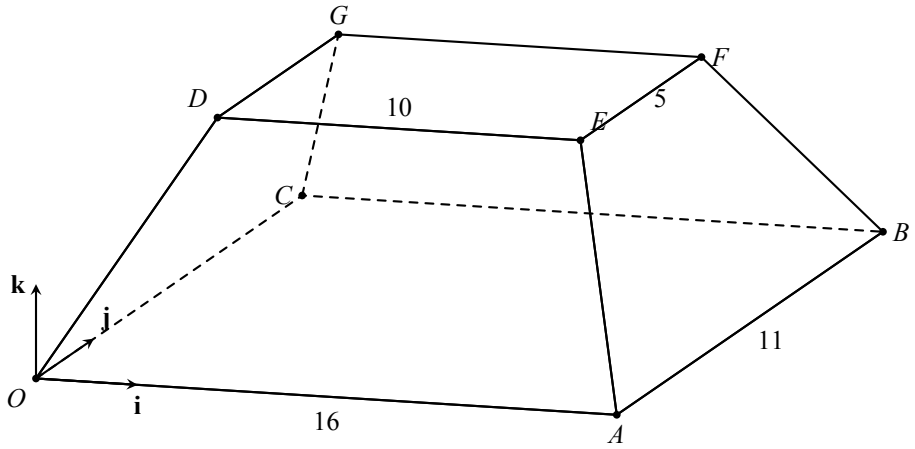
All lengths are in metres. The diagram is not drawn to scale.

Find the coordinates of all the vertices of the house $(O, B, C, D, \ldots)$. **[6]**

8

The diagram shows a container with a horizontal rectangular base $OABC$, where $OA = 16$ cm and $AB = 11$ cm. The top of the container $DEFG$ is also a horizontal rectangle, where $DE = 10$ cm and $EF = 5$ cm. The four sloping faces (for example $OAED$) of the container, each a trapezium, are inclined at the same angle to the horizontal, and the vertical distance between the base and the top is 6 cm.

The point O is taken as the origin $(0, 0, 0)$, and perpendicular unit vectors **i**, **j**, **k** are such that **i** is parallel to OA, **j** is parallel to OC, and **k** is vertical.



All lengths are in centimetres. The diagram is not drawn to scale.

Find the coordinates of D. **[3]**

Answers

1 **(a)** $\mathbf{a}=\begin{pmatrix}2\\-1\\3\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}-1\\4\\0\end{pmatrix}$, $\mathbf{a}+2\mathbf{b}=\begin{pmatrix}0\\7\\3\end{pmatrix}$

(b) $\begin{pmatrix}-7\\14\\-6\end{pmatrix}$

(c) $\begin{pmatrix}-7\\7\\-9\end{pmatrix}$

2 **(a)** $\frac{1}{7}\begin{pmatrix}2\\3\\-6\end{pmatrix}$. Its negative is also a valid answer.

(b) $\begin{pmatrix}6\\9\\-18\end{pmatrix}$. Its negative is also a valid answer.

(c) $P(5, 4, -10)$; the order is A, B, P.

3 **(a)** $k=-12,\ m=-1$

(b) Same direction; $|\mathbf{p}|:|\mathbf{q}|=3:1$

(c) $\pm\begin{pmatrix}2\\-4\\-1\end{pmatrix}$

4 **(a)** Shown. $AB:BC=1:2$

(b) $u=9,\ v=11$

5 **(a)** $\begin{pmatrix}-2\\3\\1\end{pmatrix}$

(b) $\begin{pmatrix}-8\\9\\-2\end{pmatrix}$

(c) The order is A, M, B, N; $MB:BN=1:2$

6 **(a)** $\overrightarrow{OP}=\frac{4}{7}\mathbf{a}+\frac{1}{7}\mathbf{b}$

(b) $AP:PY=3:4,\qquad BP:PX=6:1$

(c) Shown. $OP:PZ=5:2$

7 $O=(0,0,0)$, $B=(6,0,0)$, $C=(6,4,0)$, $D=(0,4,0)$, $E=(0,0,3)$, $F=(6,0,3)$, $G=(6,4,3)$, $H=(0,4,3)$, $P=(3,0,5)$, $Q=(3,4,5)$

8 $D=(3,3,6)$

END OF WORKSHEET